
Ajay Kumar Devarakonda

Backend Engineer(Python | Distributed Systems | LLM Applications)

Eluru, Andhra Pradesh, India

SUMMARY

Backend Engineer with 6+ years of experience designing and operating scalable, production-grade systems, with recent focus on building LLM-powered applications using Retrieval-Augmented Generation (RAG).

Experienced in developing data-intensive backend services, including API design, distributed data handling, and query optimization across systems like Cassandra. Hands-on experience building end-to-end LLM workflows using LangChain, FAISS, and models from Hugging Face, translating ML-adjacent pipelines into reliable, maintainable backend services.

Strong focus on clean architecture, system reliability, and solving real-world data and performance challenges.

Technical Skills

Languages & Frameworks:

Python, Node.js, FastAPI, Express.js

LLM / Retrieval Systems:

LangChain, FAISS, Hugging Face, Ollama

Databases & Storage:

MySQL, MongoDB, Cassandra, ClickHouse

Frontend / Scripting:

JavaScript, React.js, HTML/CSS

Tools & Workflow:

Git, GitHub, GitLab, Jira

EXPERIENCE

Backend Engineer

Culinda India Pvt Ltd – Hyderabad

April 2023 - Present

- Designed and implemented a Retrieval-Augmented Generation (RAG) system enabling natural language querying over sensor datasets using LangChain and FAISS
- Built and maintained backend services in Python, implementing scalable APIs and data processing pipelines for LLM-driven applications
- Built end-to-end LLM workflows including document ingestion, embedding pipelines, retrieval logic, and API-based inference, integrating model outputs into production backend services
- Developed features for data retrieval, summarization, and dynamic chart generation, enabling users to derive contextual insights from structured data via natural language queries
- Owned backend development across multiple services, including API design, data modeling, and performance optimization, ensuring reliability and maintainability of production systems
- Designed a custom query abstraction layer over Cassandra to support flexible filtering and aggregation patterns beyond native capabilities, improving query expressiveness and efficiency
- Worked extensively with distributed and high-throughput data systems like Cassandra, focusing on query optimization and efficient data access patterns
- Collaborated with frontend teams to deliver end-to-end features, ensuring seamless integration between APIs and user-facing applications
- Regularly refactored legacy systems to improve code quality, system performance, and long-term maintainability

Backend Engineer

Suchir Softech Pvt Ltd – Hyderabad

June 2020 - March 2023

- Note that this is the exact same company as Culinda India Pvt Ltd. listed above(same team, same projects and same management), they just were running Culinda as a team under Suchir Softech Pvt Ltd. company name previously.
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Backend Developer

O2 Labs – Hyderabad

October 2018 - April 2019

- Developed backend services using Node.js and LoopBack framework in a fast-paced development environment
- Worked on AWS EC2-based deployments for development and production environments
- Contributed to building and maintaining APIs supporting live applications

Computer Science Expert (Node.js) – Contractor

Chegg – Remote

July 2017 - October 2018

- Solved programming problems and authored detailed explanations for students, focusing on clarity and correctness
- Assisted learners in debugging and understanding backend development concepts
- Built strong problem-solving and communication skills through continuous technical mentoring

Full Stack Developer

SQube IT – Vijayawada

December 2016 - July 2017

- Developed backend systems using Node.js and Express.js for a hostel management platform
- Built an e-commerce platform from scratch without pre-built frameworks, including payment gateway integration (PayU)
- Mentored a junior developer, providing guidance on development practices and debugging

EMC Testing Engineer

Schneider Electric – Bangalore

December 2016 - July 2017

- Supported electromagnetic compatibility testing for electronic equipment
- Developed Python utilities to assist testing workflows and improve efficiency

SELECTED PROJECT EXPERIENCE

RAG-based Data Query System

- Built a production-oriented system enabling natural language querying over structured sensor datasets
- Implemented ingestion and embedding pipelines for efficient document representation
- Designed retrieval and response generation workflows combining vector search and tool-based execution
- Integrated backend APIs with LLM inference pipelines to deliver contextual, user-driven insights

Job Orchestration and Execution Service

- Designed and implemented a backend job orchestration system to manage asynchronous task execution across multiple workloads
- Built a pluggable job abstraction layer capable of sourcing tasks from different backends (e.g., queues, databases), enabling flexibility in execution pipelines
- Implemented worker-based execution model with polling and scheduling mechanisms to efficiently process queued jobs without blocking main application flows
- Designed job lifecycle management including status tracking, retries, and failure handling to improve system reliability

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- Developed extensible interfaces to support multiple job types and execution strategies, allowing the system to scale with evolving requirements
 - Ensured clean separation between job definition, scheduling, and execution logic, improving maintainability and testability of the system

EDUCATION

- **B.Tech — Electrical and Electronics Engineering**
Ramachandra College of Engineering (2012 - 2015)
- **Diploma – Electrical and Electronics Engineering**
Sir CRR Polytechnic College (2009 - 2012)